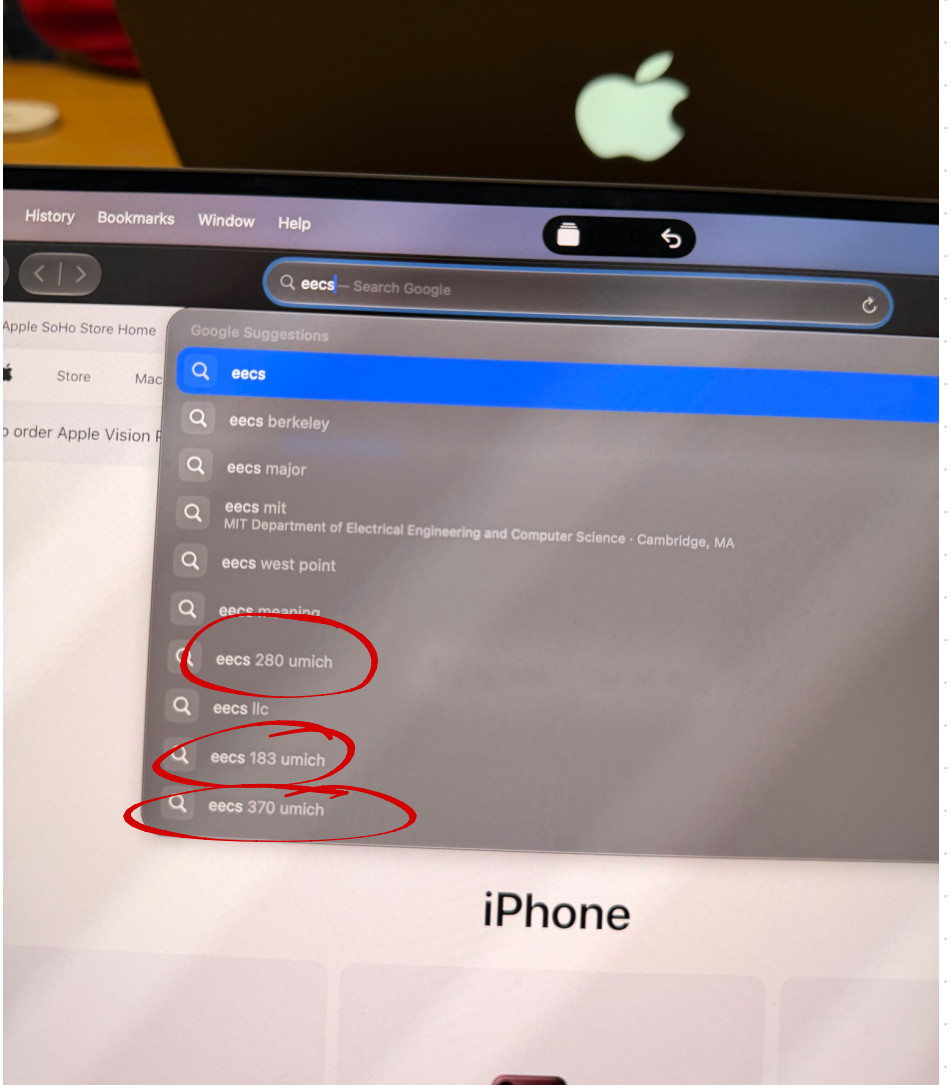
EECS 245, Fall 2026

LEC 19 Gradient Descent and Convexity

→ Read: Ch. 8.4 - 8.5


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Agenda

Today's content is in scope, Thursday's is not!

- Recap: gradient descent
- Gradient descent for linear regression
- Convexity
  - Formal definition
  - Local and global minima
  - Relationship to gradient descent

Announcements

- HW 8 due Sunday; no slip days
- Midterm 2 on Tuesday, November 17th, 7-9PM, 1013 DOW
  - → Everything since MT1, including today
  - → 2 handwritten notes sheets
  - → See past exams at exams.eecs245.org
  - → Details on Ed

Goal of gradient descent: minimize $f(\vec{x})$,

where $f: \mathbb{R}^d \to \mathbb{R}$,

given $\nabla f(\vec{x})$: assume we can find $\nabla f(\vec{x})$

but solving $\nabla f(\vec{x}) = \vec{0}$ is hard

Gradient descent update rule:

$$\vec{x}^{(t+1)} = \vec{x}^{(t)} - \alpha \nabla f(\vec{x}^{(t)})$$

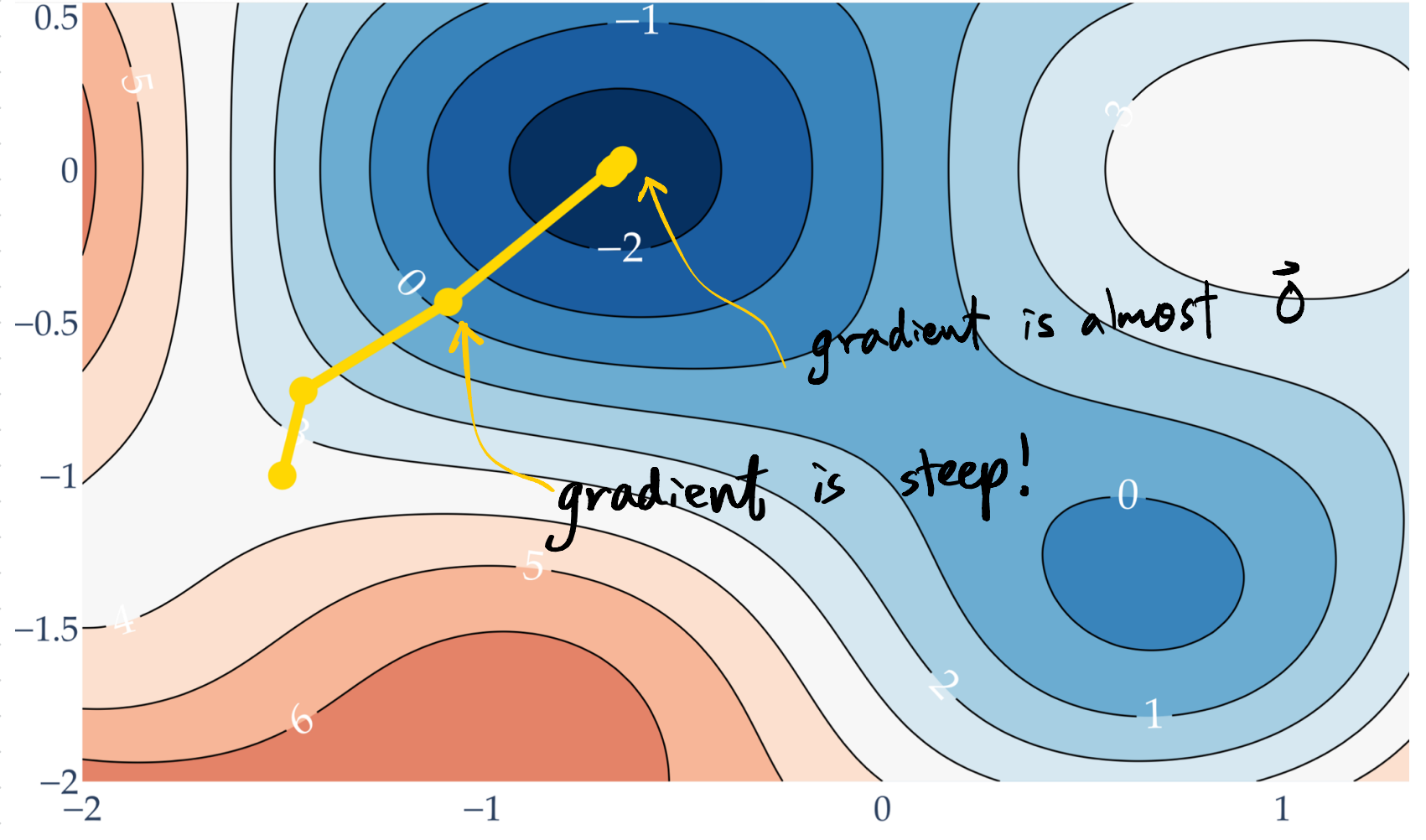
gradient at old guess

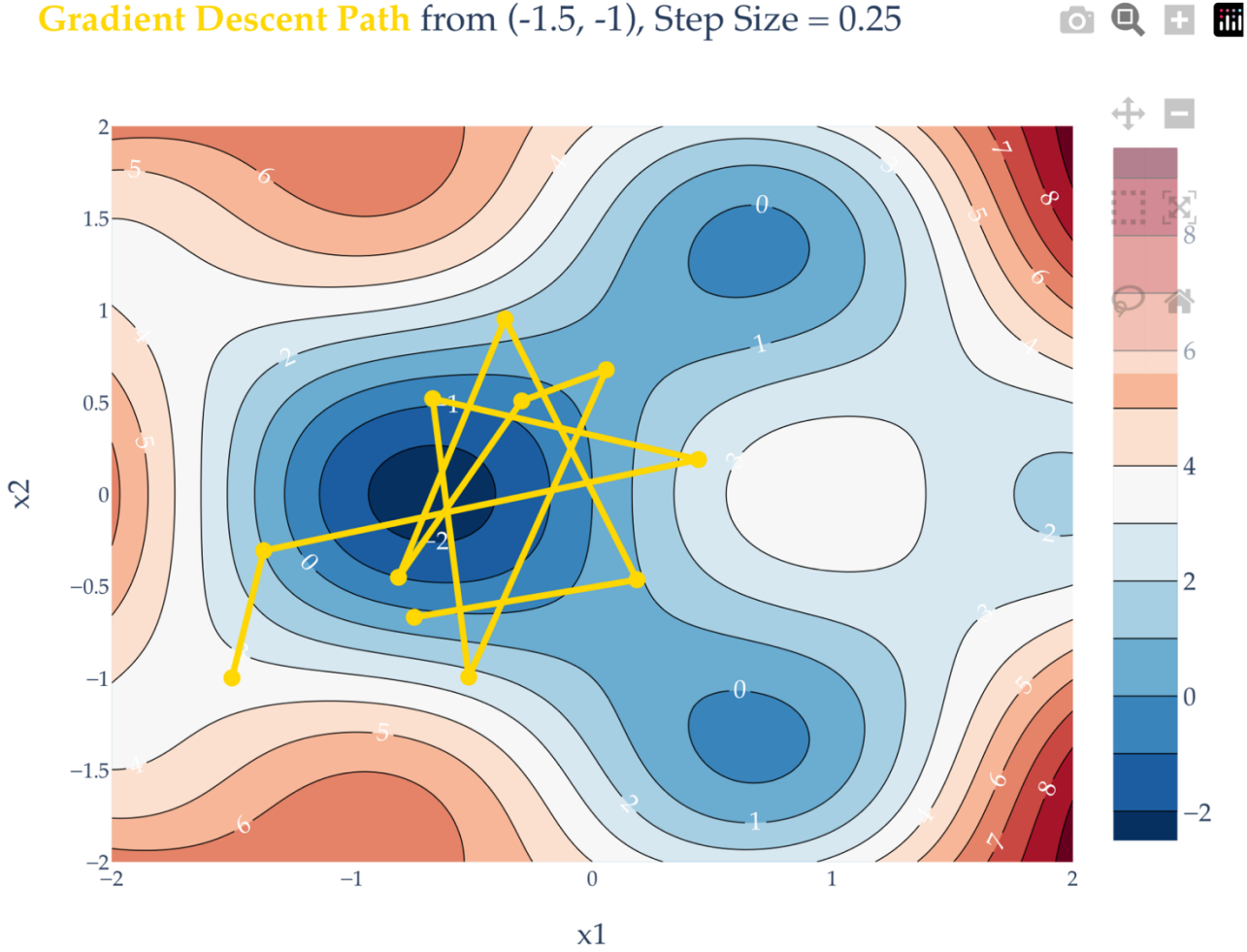
new guess | old guess | learning rate / step size

Ahead of time, choose α, $\vec{x}^{(0)}$: initial guess

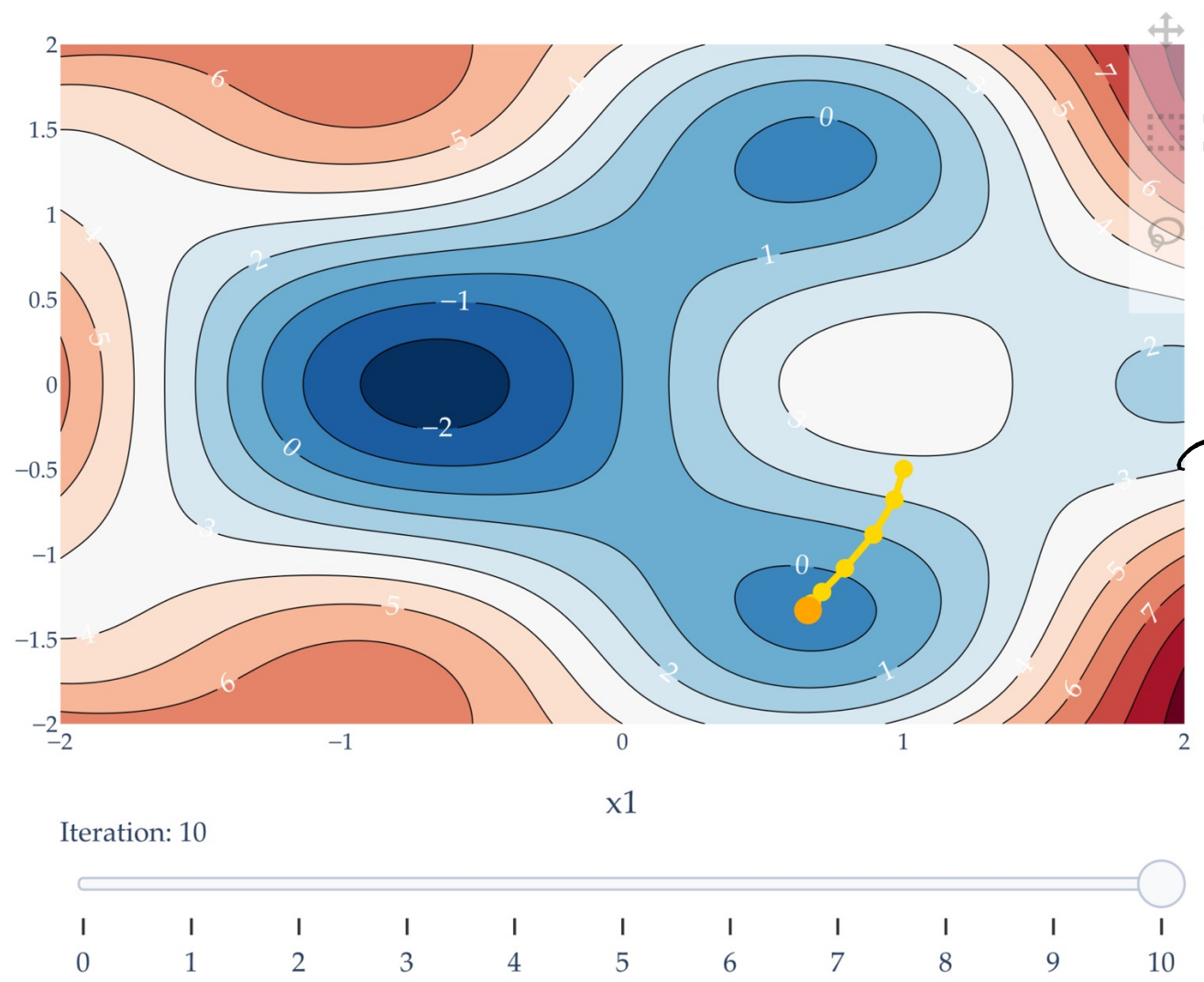
terminate when $\|\nabla f(\vec{x}^{(t)})\| \leq 0.001$

10:51AM Tue Nov 10 Safari
notes.eecs245.org





Same $\vec{x}^{(0)}$ as before, but α too big!



different $\vec{x}^{(0)}$: converged at local minimum: "trap"

Gradient descent for multiple linear regression

$$R_{sq}(\vec{w}) = \frac{1}{n} \|\vec{y} - X\vec{w}\|^2$$

$\vec{y}$: observation vec

X: design matrix

We know $\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$

$\vec{w}^*$: optimal parameters

$$(X^T X)\vec{w} = X^T \vec{y}$$

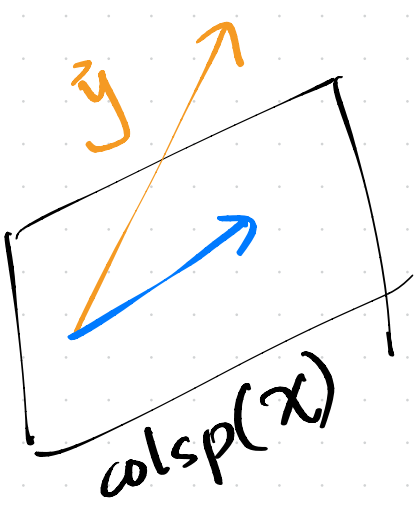
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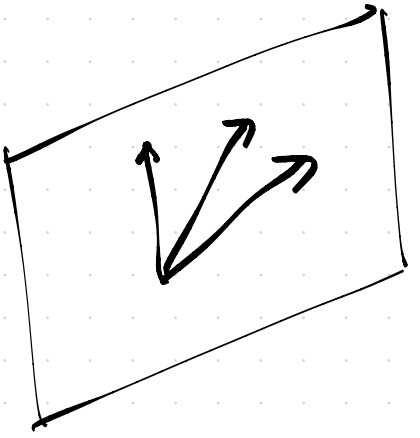
still could use gradient descent to estimate $\vec{w}^*$

$$\nabla R_{sq}(\vec{w}) = \frac{2}{n}\left(X^T X \vec{w} - X^T \vec{y}\right)$$

$$\vec{w}^{(t+1)} = \vec{w}^{(t)} - \alpha \frac{2}{n} \left(X^T X \vec{w}^{(t)} - X^T \vec{y} \right)$$

→ Given $\vec{w}^{(0)}$ and α, we can use GD to find $\vec{w}^*$!


$\vec{y}$
colsp(X)



approach w_0^*, w_1^*

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```
| w^(0) = [   0.000,    0.000] | mse = 5523.5231 | ||grad|| = 1229.842640
| w^(500) = [  19.804,    6.102] | mse =  314.8511 | ||grad|| =  3.527944
| w^(1000) = [  36.133,    4.200] | mse =  200.7135 | ||grad|| =  3.058219
| w^(1500) = [  50.289,    2.551] | mse =  220.0323 | ||grad|| =  2.651035
| w^(2000) = [  62.559,    1.121] | mse =  189.4630 | ||grad|| =  2.298065
| w^(2500) = [  73.196,   -0.118] | mse =  166.4919 | ||grad|| =  1.992091
| w^(3000) = [  82.416,   -1.193] | mse =  149.2306 | ||grad|| =  1.726856
| w^(3500) = [  90.409,   -2.124] | mse =  136.2598 | ||grad|| =  1.496935
| w^(4000) = [  97.338,   -2.931] | mse =  126.5130 | ||grad|| =  1.297627
| w^(4500) = [ 103.344,   -3.631] | mse =  119.1889 | ||grad|| =  1.124856
| w^(5000) = [ 108.551,   -4.237] | mse =  113.6852 | ||grad|| =  0.975088
| w^(5500) = [ 113.064,   -4.763] | mse =  109.5496 | ||grad|| =  0.845261
| w^(6000) = [ 116.976,   -5.219] | mse =  106.4419 | ||grad|| =  0.732719
| w^(6500) = [ 120.368,   -5.614] | mse =  104.1067 | ||grad|| =  0.635162
| w^(7000) = [ 123.308,   -5.957] | mse =  102.3519 | ||grad|| =  0.550594
| w^(7500) = [ 125.856,   -6.254] | mse =  101.0333 | ||grad|| =  0.477285
| w^(8000) = [ 128.065,   -6.511] | mse =  100.0424 | ||grad|| =  0.413738
| w^(8500) = [ 129.980,   -6.734] | mse =   99.2978 | ||grad|| =  0.358651
| w^(9000) = [ 131.640,   -6.928] | mse =   98.7383 | ||grad|| =  0.310899
| w^(9500) = [ 133.079,   -7.095] | mse =   98.3179 | ||grad|| =  0.269504
| w^(10000) = [ 134.327,   -7.241] | mse =   98.0020 | ||grad|| =  0.233621
| w^(10500) = [ 135.408,   -7.367] | mse =   97.7646 | ||grad|| =  0.202516
| w^(11000) = [ 136.345,   -7.476] | mse =   97.5862 | ||grad|| =  0.175552
| w^(11500) = [ 137.158,   -7.571] | mse =   97.4521 | ||grad|| =  0.152179
| w^(12000) = [ 137.862,   -7.653] | mse =   97.3514 | ||grad|| =  0.131917
| w^(12500) = [ 138.473,   -7.724] | mse =   97.2757 | ||grad|| =  0.114353
```

CONTENTS

Simple Linear Regression

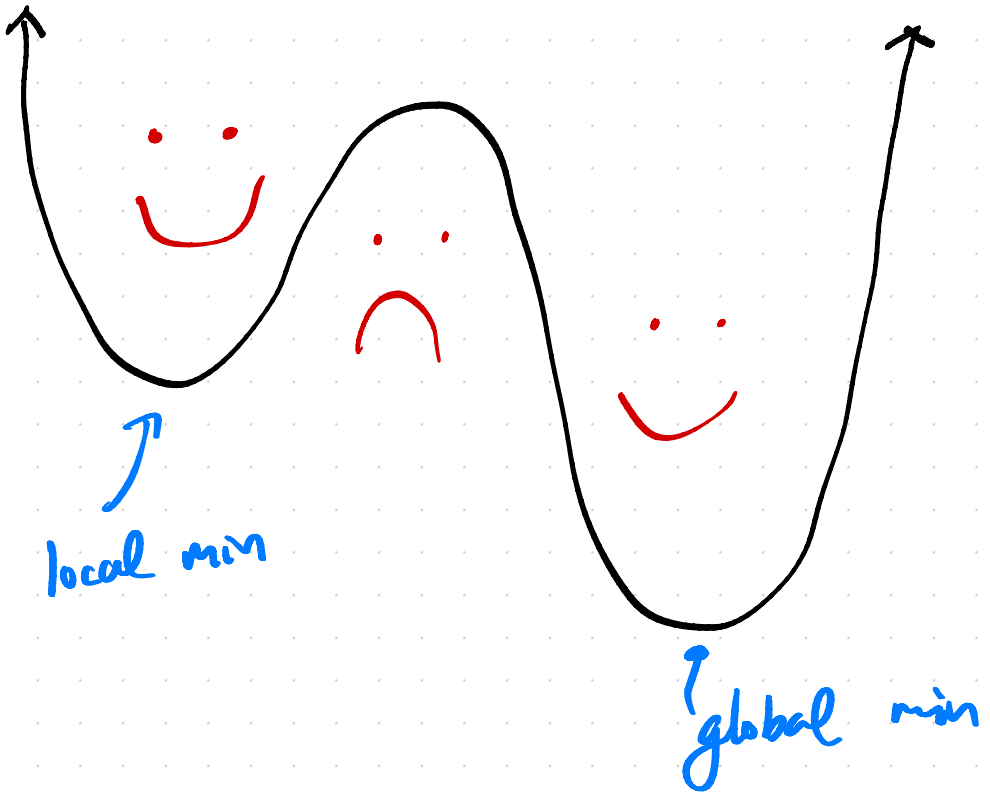
Another Example: Logistic Regression

Issues of Scale

MSE approaching the theoretical minimum

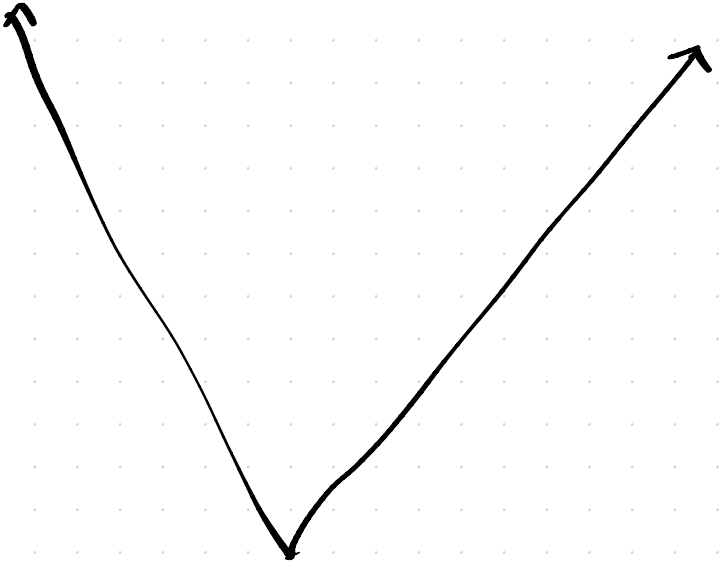
$\|\nabla R_{sq}(\vec{w}^{(t)})\| \to 0$

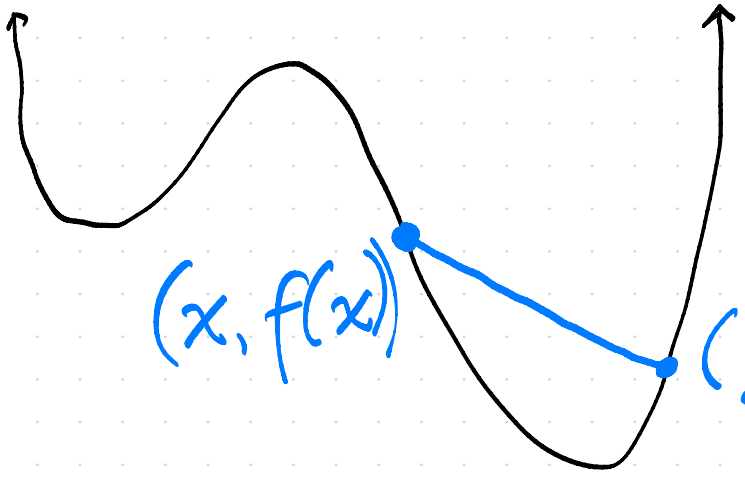
attendance, 11:19



convex ?

no, but convex in some intervals



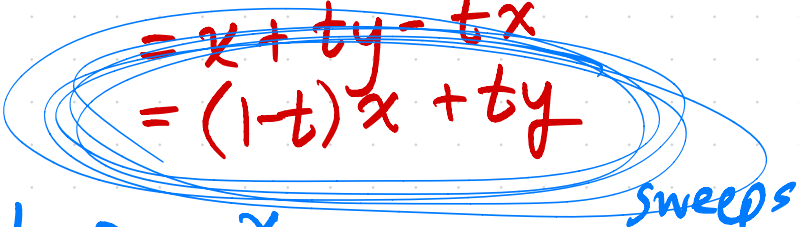


Idea: f convex if every secant line lies on or above the function.

$$\underbrace{f\big((1-t)x + ty\big)}_{\text{function}} \leq \underbrace{(1-t)f(x) + tf(y)}_{\text{line}}$$

Aside: what number is t% of the way from x to y?

$$x + t(y-x)$$
$$= x + ty - tx$$
$$= (1-t)x + ty$$

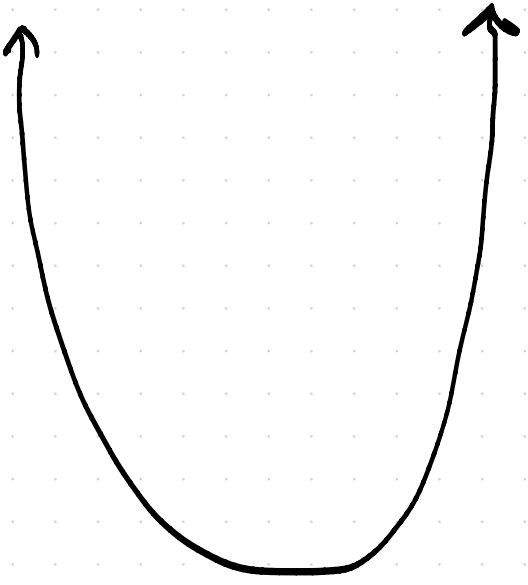


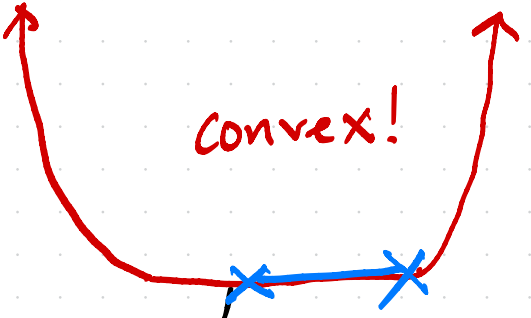
sweeps from x to y

if $t=0$, x

$t=1$, y

$t=1/2$, $\frac{x+y}{2}$





convex!

$f(x) = e^x$

Convexity $f : \mathbb{R}^d \to \mathbb{R}$ is convex if

$$f\big((1-t)\vec{x} + t\vec{y}\big) \leq (1-t) f(\vec{x}) + t f(\vec{y})$$

function — secant line

for all $\vec{x}, \vec{y} \in$ domain,

$$0 \leq t \leq 1$$

---

strict convexity: $<$ instead of $\leq$,

$$0 < t < 1$$

Connection to gradient descent?

big idea: if f convex, then any local min is a global min!

→ convex functions don't have "traps"!

proof by contradiction: suppose $\vec{x}^*$ is a local min, but $\vec{z} \neq \vec{x}^*$ is global min

$$f(\vec{z}) < f(\vec{x}^*)$$

convexity: $f((1-t)\vec{x}^* + t\vec{z}) \leq (1-t)f(\vec{x}^*) + tf(\vec{z})$

$$< (1-t)f(\vec{x}^*) + tf(\vec{x}^*)$$

convexity: $f((1-t)\vec{x}^* + t\vec{z}) \leq (1-t)f(\vec{x}^*) + tf(\vec{z})$

$< (1-t)f(\vec{x}^*) + tf(\vec{x}^*)$

$= f(\vec{x}^*)$

says

$$f((1-t)\vec{x}^* + t\vec{z}) < f(\vec{x}^*)$$

for all $t \in [0,1]$

if $t = 0.0001$, this gives me an input $\vec{i}$ really close to $\vec{x}^*$

with $f(\vec{i}) < f(\vec{x}^*)$: contradiction, because $\vec{x}^*$ is a local min!